

Value of Information Analysis

METIMMOX-1 Economic Evaluation

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Table of contents

1	Overview	2
2	Model Structure Definitions	3
3	Scenario Definitions	4
4	Base Case Scenario	5
4.1	EVPI Comparison Across Model Structures	5
4.2	EVPPPI Results by Model Structure	5
5	Scenario 1: Decreased Nivolumab Cost	7
5.1	EVPI Comparison Across Model Structures	7
5.2	EVPPPI Results by Model Structure	7
6	Scenario 2: Higher WTP Threshold (EUR100,000)	9
6.1	EVPI Comparison Across Model Structures	9
6.2	EVPPPI Results by Model Structure	9
7	Scenario 3: Higher WTP with Decreased Nivolumab Cost	11
7.1	EVPI Comparison Across Model Structures	11
7.2	EVPPPI Results by Model Structure	11
8	Scenario 4: Highest WTP Threshold (EUR150,000)	13
8.1	EVPI Comparison Across Model Structures	13
8.2	EVPPPI Results by Model Structure	13
9	Scenario 5: Highest WTP with Decreased Nivolumab Cost	15
9.1	EVPI Comparison Across Model Structures	15
9.2	EVPPPI Results by Model Structure	15
10	Comparative Analysis Across Scenarios and Models	17
10.1	EVPI Comparison	17
10.2	EVPPPI Parameter Heatmap by Model Structure	18
10.3	Summary Comparison Table	18
10.4	Key Findings	19
10.4.1	Model Structure Robustness	19
10.4.2	Scenario with Greatest Decision Uncertainty	19
10.4.3	Most Important Parameter Across All Scenarios	19

1 Overview

This report presents the results of an Expected Value of Partially Perfect Information (EVPPI) analysis for the METIMMOX-1 decision-analytic model. The analysis evaluates the value of reducing uncertainty in individual model parameters across multiple scenarios with varying willingness-to-pay (WTP) thresholds and nivolumab drug costs.

Key Feature: This report compares EVPI and EVPPI results across three different model structures (Model A, B, C) to assess the robustness of value of information findings to modeling assumptions about how biomarker interactions are specified.

The Expected Value of Partially Perfect Information (EVPPI) quantifies the value of eliminating uncertainty in specific model parameters. It represents the maximum amount that should be paid for additional research to perfectly inform a particular parameter. EVPI (Expected Value of Perfect Information) represents the value of eliminating all uncertainty, and EVPPI for any parameter cannot exceed total EVPI.

2 Model Structure Definitions

We compare three model structures to assess the robustness of value of information results to the specification of biomarker-treatment interactions in the survival models.

Table 1: Model structure definitions used in the analysis

Model	Name	Formula Pattern	Description
Model A	Joint	Age + sex + Rx + crp:Rx + tlr:Rx + tmb_braf:Rx	All biomarkers and all treatment interactions in one unified model
Model B	Focused	Age + sex + Rx + crp + tlr + tmb_braf + [biomarker]:Rx	All biomarkers as main effects, with only one interaction per biomarker-specific model
Model C	Separate	Age + sex + Rx + [biomarker]:Rx	Only one biomarker and its treatment interaction per model (simplest specification)

Rationale: Model A (Joint) is the most comprehensive specification but may suffer from over-fitting with limited sample size. Model C (Separate) is the simplest but ignores potential confounding from other biomarkers. Model B (Focused) represents a middle ground. Comparing results across all three structures provides insight into the structural uncertainty of our findings.

3 Scenario Definitions

Table 2: Analysis scenario definitions. Higher WTP thresholds (EUR100,000 and EUR150,000) are arbitrary exploratory thresholds, though the highest severity adjustment in Norwegian guidelines may fall between these values. The decreased nivolumab cost (EUR4,641) represents an 80% discount compared to the 40% discount in the base case scenario.

Scenario Name	WTP Threshold	Nivolumab Cost
Base Case	EUR 51,000	EUR13,923
Decreased Nivolumab Cost	EUR 51,000	EUR 4,641
WTP €100,000	EUR100,000	EUR13,923
WTP €100,000 + Decreased Nivolumab	EUR100,000	EUR 4,641
WTP €150,000	EUR150,000	EUR13,923
WTP €150,000 + Decreased Nivolumab	EUR150,000	EUR 4,641

4 Base Case Scenario

Willingness-to-Pay Threshold: EUR51,000

Nivolumab Cost: EUR13,923

4.1 EVPI Comparison Across Model Structures

Table 3: Expected Value of Perfect Information (EVPI) by model structure - Base Case

	Model	EVPI
5	Model A: Joint	EUR1291.45
56	Model B: Focused	EUR103.98
57	Model C: Separate	EUR940.43

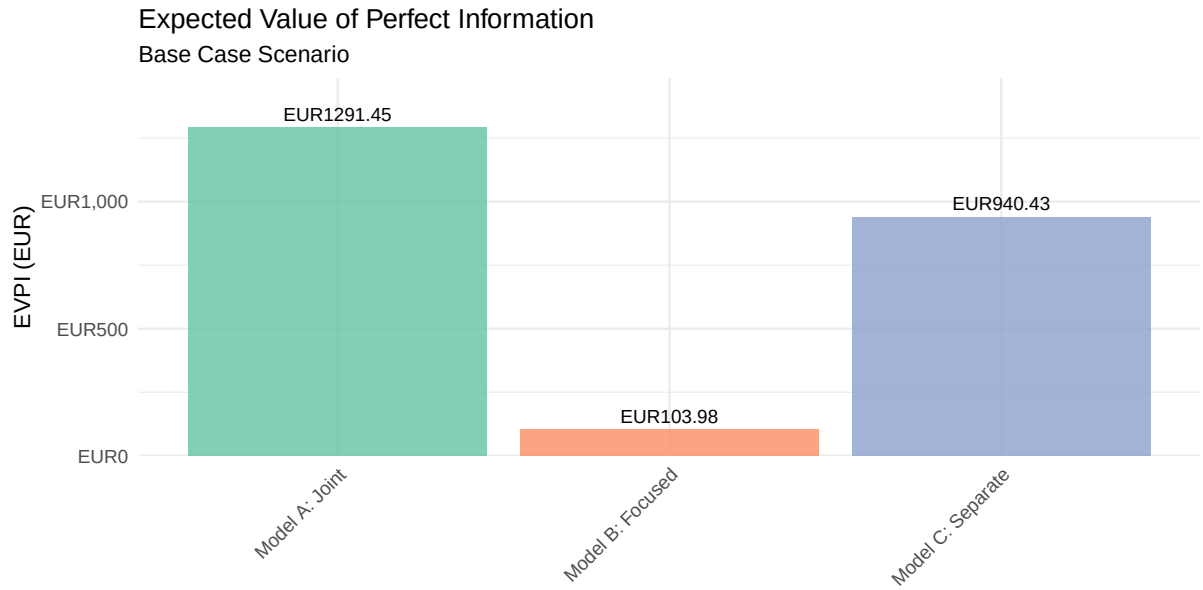


Figure 1: EVPI comparison across model structures (Base Case)

4.2 EVPPI Results by Model Structure

Table 4: EVPPI results by model structure - Base Case (ranked by average importance)

Parameter	EVPI (A)	% (A)	EVPI (B)	% (B)	EVPI (C)	% (C)
u_np	EUR903.95	70%	EUR103.98	100%	EUR828.86	88.1%
u_p	EUR786.79	60.9%	EUR103.98	100%	EUR796.63	84.7%
c_other_last	EUR99.77	7.7%	EUR103.98	100%	EUR107.85	11.5%
c_test_NGS	EUR82.23	6.4%	EUR90.12	86.7%	EUR109.53	11.6%
c_other_baseline	EUR0	0%	EUR103.98	100%	EUR40.94	4.4%
c_test_CT	EUR30.73	2.4%	EUR103.66	99.7%	EUR0	0%
c_other_follow	EUR119.23	9.2%	EUR39.55	38%	EUR0	0%
c_drug_FLOX	EUR80.65	6.2%	EUR28.14	27.1%	EUR42.34	4.5%
c_drug_nivo	EUR76.3	5.9%	EUR11.47	11%	EUR0	0%
c_other_visit	EUR0	0%	EUR43.06	41.4%	EUR0	0%
c_test_blood	EUR25.97	2%	EUR0	0%	EUR0	0%

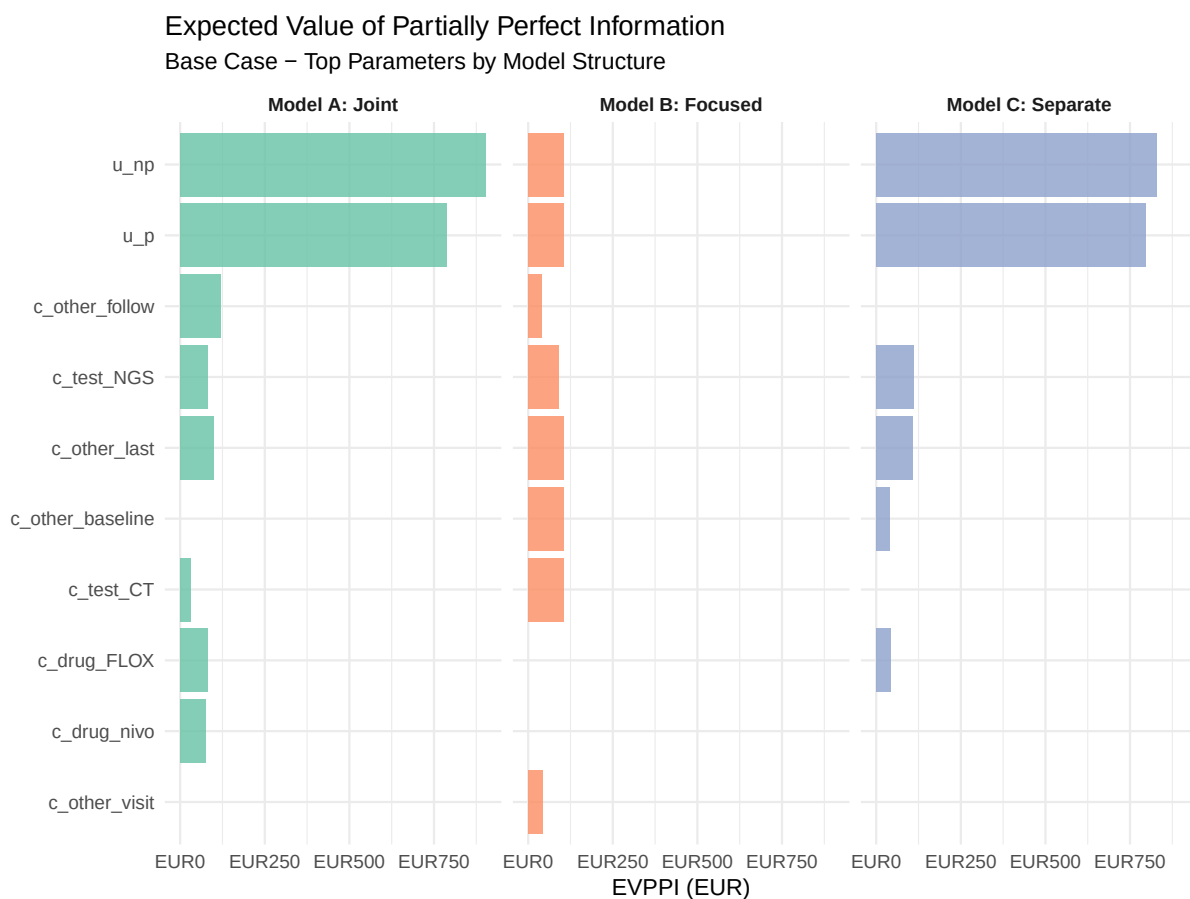


Figure 2: EVVPI by parameter and model structure (Base Case)

The positive EVVPI indicates decision uncertainty at the base case WTP threshold. This suggests that additional research to reduce parameter uncertainty could have value. The comparison across model structures shows whether this finding is robust to the specification of biomarker-treatment interactions.

5 Scenario 1: Decreased Nivolumab Cost

Willingness-to-Pay Threshold: EUR51,000

Nivolumab Cost: EUR4,641 (80% discount)

5.1 EVPI Comparison Across Model Structures

Table 5: Expected Value of Perfect Information (EVPI) by model structure - Scenario 1

	Model	EVPI
51	Model A: Joint	EUR8728.45
611	Model B: Focused	EUR2235.35
512	Model C: Separate	EUR8326.24

5.2 EVPPI Results by Model Structure

Table 6: EVPPI results by model structure - Scenario 1: Decreased Nivolumab Cost

Parameter	EVPI (A)	% (A)	EVPI (B)	% (B)	EVPI (C)	% (C)
u_p	EUR760.36	8.7%	EUR1011.52	45.3%	EUR765.81	9.2%
u_np	EUR813.71	9.3%	EUR845.42	37.8%	EUR855.66	10.3%
c_test_blood	EUR0	0%	EUR131.76	5.9%	EUR88.42	1.1%
c_other_last	EUR92.1	1.1%	EUR0	0%	EUR123.99	1.5%
c_other_baseline	EUR20.32	0.2%	EUR54.47	2.4%	EUR31.05	0.4%
c_other_follow	EUR23.44	0.3%	EUR29.5	1.3%	EUR41.75	0.5%
c_drug_nivo	EUR0	0%	EUR22.14	1%	EUR56.56	0.7%
c_test_CT	EUR0	0%	EUR14.53	0.7%	EUR61.32	0.7%
c_other_visit	EUR0	0%	EUR30.64	1.4%	EUR21.28	0.3%
c_test_NGS	EUR16.46	0.2%	EUR0	0%	EUR29.7	0.4%
c_drug_FLOX	EUR0	0%	EUR0	0%	EUR0	0%

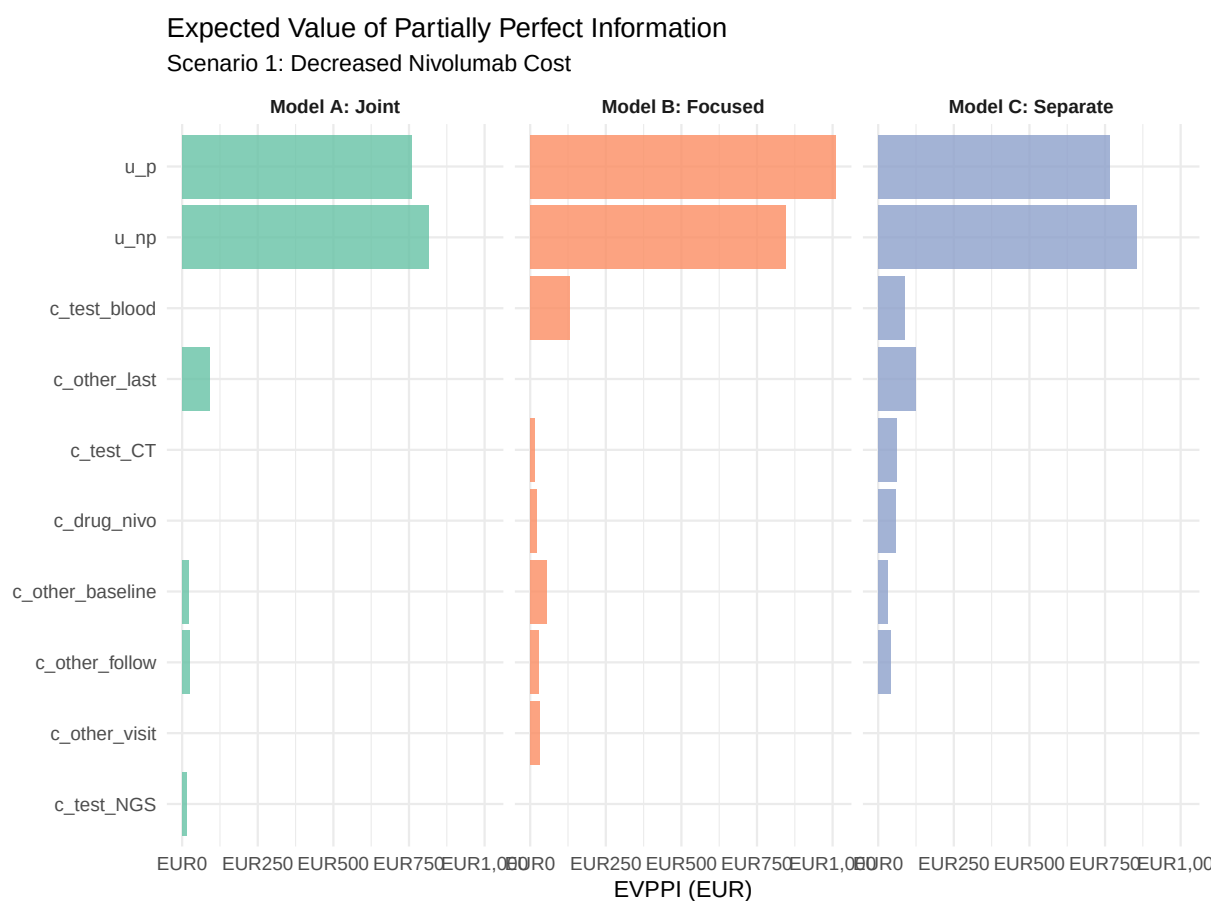


Figure 3: EVPPPI by parameter and model structure (Scenario 1)

6 Scenario 2: Higher WTP Threshold (EUR100,000)

Willingness-to-Pay Threshold: EUR1e+05

Nivolumab Cost: EUR13,923 (standard)

6.1 EVPI Comparison Across Model Structures

Table 7: Expected Value of Perfect Information (EVPI) by model structure - Scenario 2

	Model	EVPI
52	Model A: Joint	EUR11036.3
521	Model B: Focused	EUR2233.74
522	Model C: Separate	EUR10320.09

6.2 EVPPI Results by Model Structure

Table 8: EVPPI results by model structure - Scenario 2: WTP EUR100,000

Parameter	EVPPI (A)	% (A)	EVPPI (B)	% (B)	EVPPI (C)	% (C)
u_np	EUR1959.34	17.8%	EUR1976.48	88.5%	EUR1722.5	16.7%
u_p	EUR1632.16	14.8%	EUR1951.51	87.4%	EUR1542.8	14.9%
c_drug_FLOX	EUR155.79	1.4%	EUR61.49	2.8%	EUR1.03	0%
c_other_baseline	EUR0	0%	EUR0	0%	EUR114.62	1.1%
c_other_follow	EUR0	0%	EUR0	0%	EUR100.25	1%
c_drug_nivo	EUR0	0%	EUR0	0%	EUR98.37	1%
c_test_NGS	EUR62.25	0.6%	EUR0	0%	EUR23.03	0.2%
c_other_last	EUR84.52	0.8%	EUR0	0%	EUR0	0%
c_other_visit	EUR0	0%	EUR77.66	3.5%	EUR0	0%
c_test_blood	EUR0	0%	EUR0	0%	EUR39.79	0.4%
c_test_CT	EUR0	0%	EUR0	0%	EUR0	0%

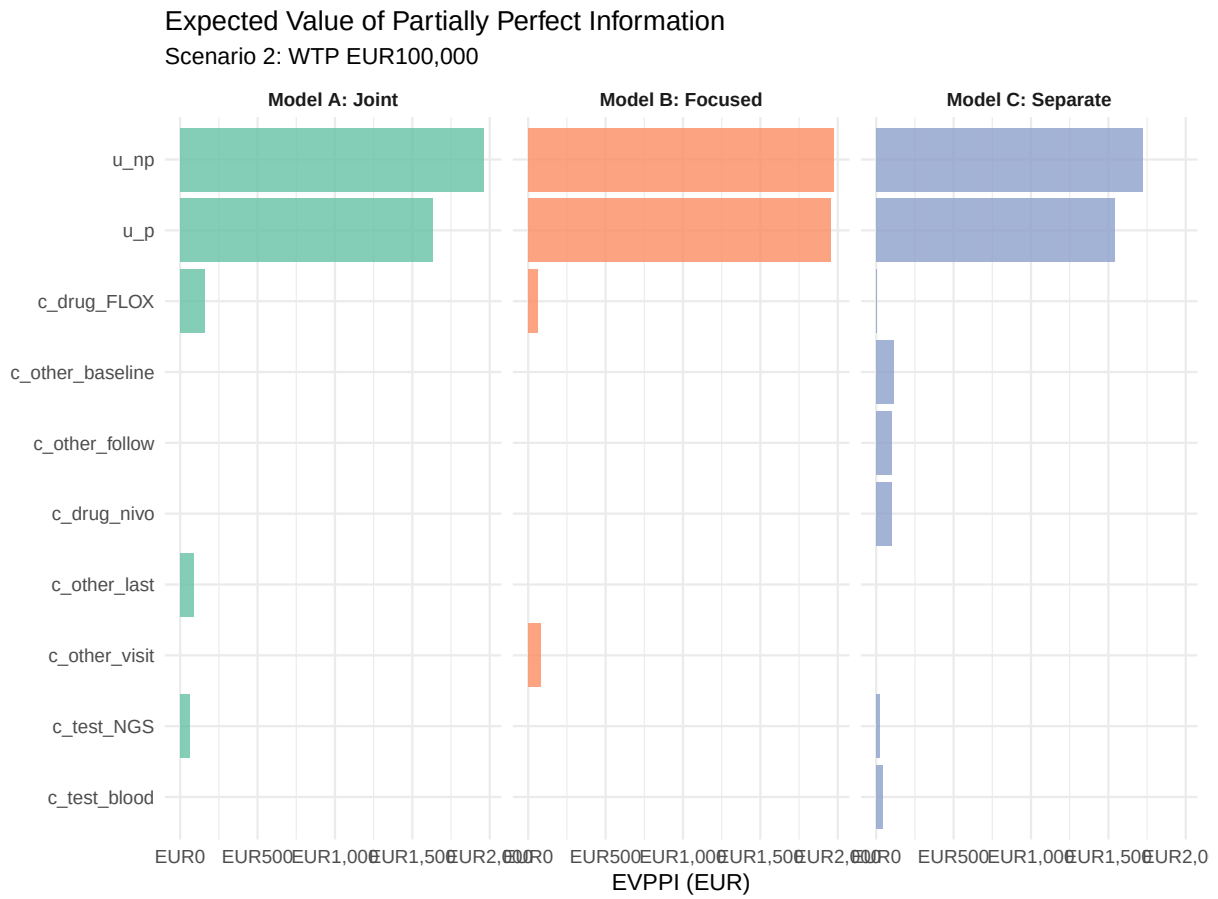


Figure 4: EVPPPI by parameter and model structure (Scenario 2)

7 Scenario 3: Higher WTP with Decreased Nivolumab Cost

Willingness-to-Pay Threshold: EUR1e+05

Nivolumab Cost: EUR4,641 (80% discount)

7.1 EVPI Comparison Across Model Structures

Table 9: Expected Value of Perfect Information (EVPI) by model structure - Scenario 3

	Model	EVPI
53	Model A: Joint	EUR17969.61
531	Model B: Focused	EUR8134.99
532	Model C: Separate	EUR18313.84

7.2 EVPPI Results by Model Structure

Table 10: EVPPI results by model structure - Scenario 3: WTP EUR100,000 + Decreased Nivolumab

Parameter	EVPI (A)	% (A)	EVPI (B)	% (B)	EVPI (C)	% (C)
u_np	EUR2159.89	12%	EUR2210.46	27.2%	EUR2637.27	14.4%
u_p	EUR1715.38	9.5%	EUR1622.96	20%	EUR1613.81	8.8%
c_other_last	EUR293.72	1.6%	EUR111.23	1.4%	EUR42.96	0.2%
c_other_visit	EUR0	0%	EUR0	0%	EUR193.45	1.1%
c_test_blood	EUR164.55	0.9%	EUR0	0%	EUR24.31	0.1%
c_test_NGS	EUR0	0%	EUR51.09	0.6%	EUR74.8	0.4%
c_drug_nivo	EUR8.59	0%	EUR0	0%	EUR113.68	0.6%
c_other_baseline	EUR89.66	0.5%	EUR0	0%	EUR0	0%
c_drug_FLOX	EUR0	0%	EUR6.48	0.1%	EUR65.18	0.4%
c_test_CT	EUR0	0%	EUR0	0%	EUR58.13	0.3%
c_other_follow	EUR0	0%	EUR0	0%	EUR0	0%

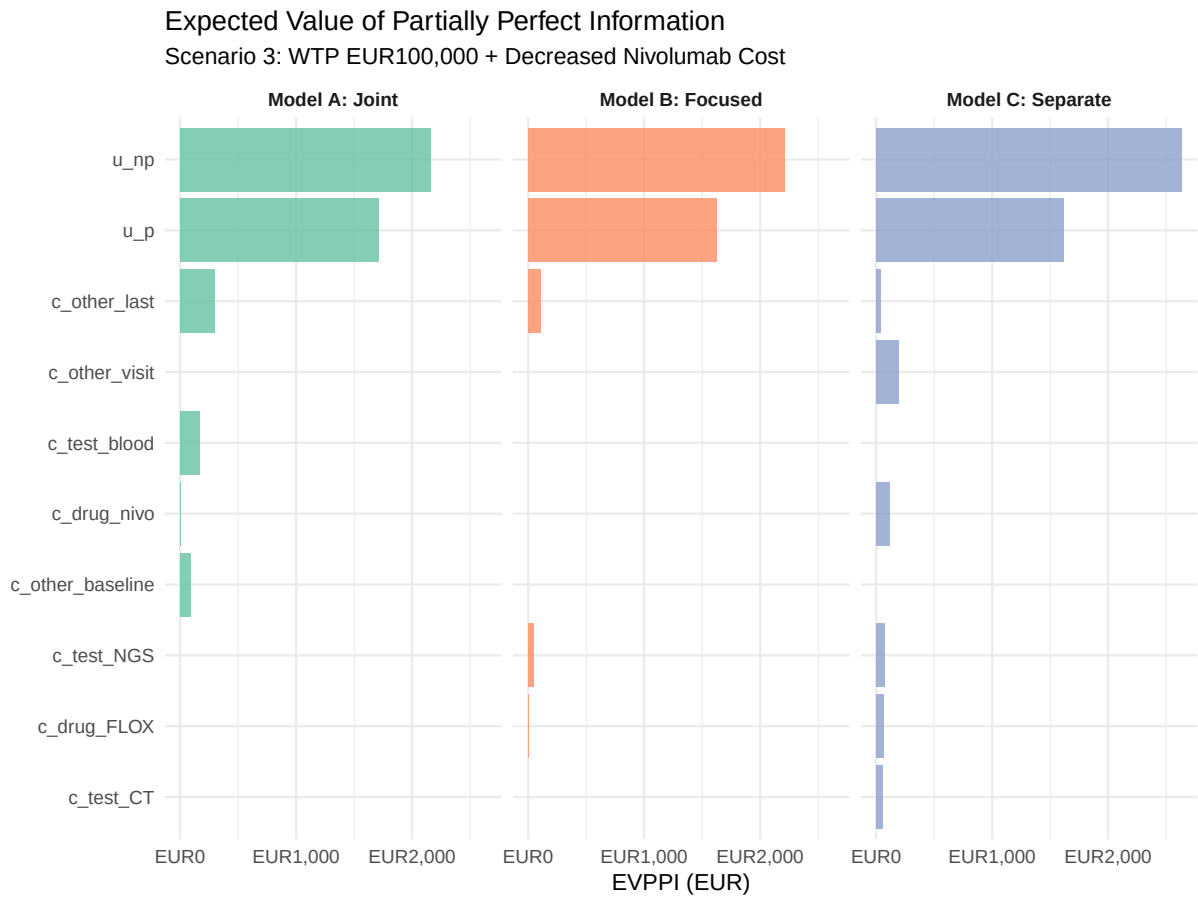


Figure 5: EVPPI by parameter and model structure (Scenario 3)

8 Scenario 4: Highest WTP Threshold (EUR150,000)

Willingness-to-Pay Threshold: EUR150,000

Nivolumab Cost: EUR13,923 (standard)

8.1 EVPI Comparison Across Model Structures

Table 11: Expected Value of Perfect Information (EVPI) by model structure - Scenario 4

	Model	EVPI
54	Model A: Joint	EUR26232
541	Model B: Focused	EUR6287.53
542	Model C: Separate	EUR25542.16

8.2 EVPPI Results by Model Structure

Table 12: EVPPI results by model structure - Scenario 4: WTP EUR150,000

Parameter	EVPPI (A)	% (A)	EVPPI (B)	% (B)	EVPPI (C)	% (C)
u_np	EUR2673.87	10.2%	EUR3100.7	49.3%	EUR2616.38	10.2%
u_p	EUR2214.72	8.4%	EUR2900.92	46.1%	EUR2351.83	9.2%
c_drug_nivo	EUR645.73	2.5%	EUR70.09	1.1%	EUR582.13	2.3%
c_other_last	EUR359.14	1.4%	EUR307.65	4.9%	EUR349.31	1.4%
c_test_blood	EUR17.89	0.1%	EUR329.26	5.2%	EUR141.73	0.6%
c_other_follow	EUR0	0%	EUR111.7	1.8%	EUR199.14	0.8%
c_other_visit	EUR7.61	0%	EUR138.41	2.2%	EUR106.11	0.4%
c_other_baseline	EUR224.61	0.9%	EUR0	0%	EUR15.01	0.1%
c_test_CT	EUR134.28	0.5%	EUR90.76	1.4%	EUR0	0%
c_drug_FLOX	EUR66.19	0.3%	EUR0	0%	EUR94.17	0.4%
c_test_NGS	EUR92.59	0.4%	EUR0	0%	EUR62.64	0.2%

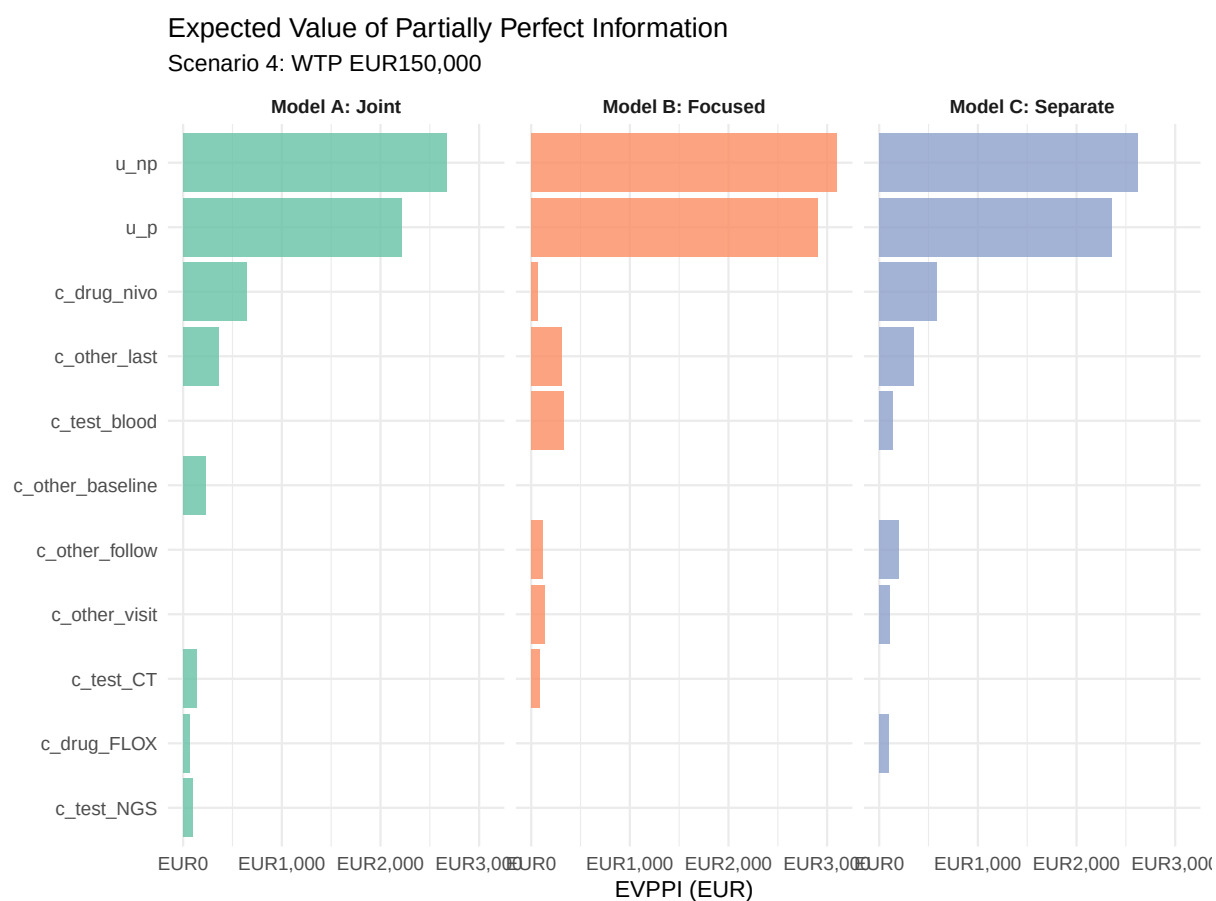


Figure 6: EVVPI by parameter and model structure (Scenario 4)

9 Scenario 5: Highest WTP with Decreased Nivolumab Cost

Willingness-to-Pay Threshold: EUR150,000

Nivolumab Cost: EUR4,641 (80% discount)

9.1 EVPI Comparison Across Model Structures

Table 13: Expected Value of Perfect Information (EVPI) by model structure - Scenario 5

	Model	EVPI
55	Model A: Joint	EUR25162.41
551	Model B: Focused	EUR14756.5
552	Model C: Separate	EUR25182.37

9.2 EVPPI Results by Model Structure

Table 14: EVPPI results by model structure - Scenario 5: WTP EUR150,000 + Decreased Nivolumab

Parameter	EVPI (A)	% (A)	EVPI (B)	% (B)	EVPI (C)	% (C)
u_np	EUR2914.08	11.6%	EUR3317.74	22.5%	EUR3410.67	13.5%
u_p	EUR2581.63	10.3%	EUR2794.13	18.9%	EUR2042.8	8.1%
c_drug_FLOX	EUR203.81	0.8%	EUR0	0%	EUR195.56	0.8%
c_test_NGS	EUR274.88	1.1%	EUR0	0%	EUR108.23	0.4%
c_drug_nivo	EUR149.19	0.6%	EUR69.26	0.5%	EUR0	0%
c_test_CT	EUR44.74	0.2%	EUR16.33	0.1%	EUR98.44	0.4%
c_other_last	EUR0	0%	EUR0	0%	EUR143.27	0.6%
c_other_baseline	EUR0	0%	EUR43.85	0.3%	EUR0	0%
c_other_visit	EUR9.62	0%	EUR0	0%	EUR0	0%
c_test_blood	EUR0	0%	EUR0	0%	EUR0	0%
c_other_follow	EUR0	0%	EUR0	0%	EUR0	0%

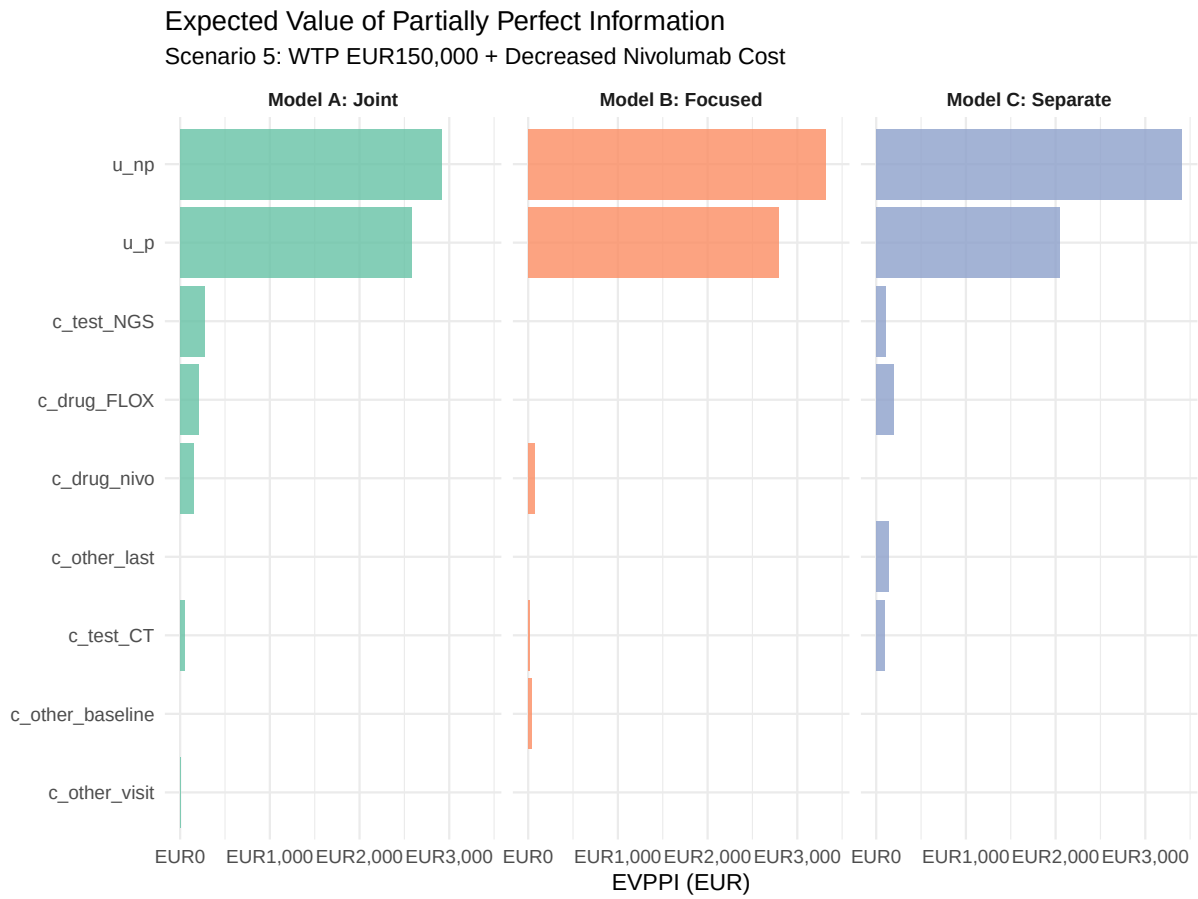


Figure 7: EVPPI by parameter and model structure (Scenario 5)

10 Comparative Analysis Across Scenarios and Models

10.1 EVPI Comparison

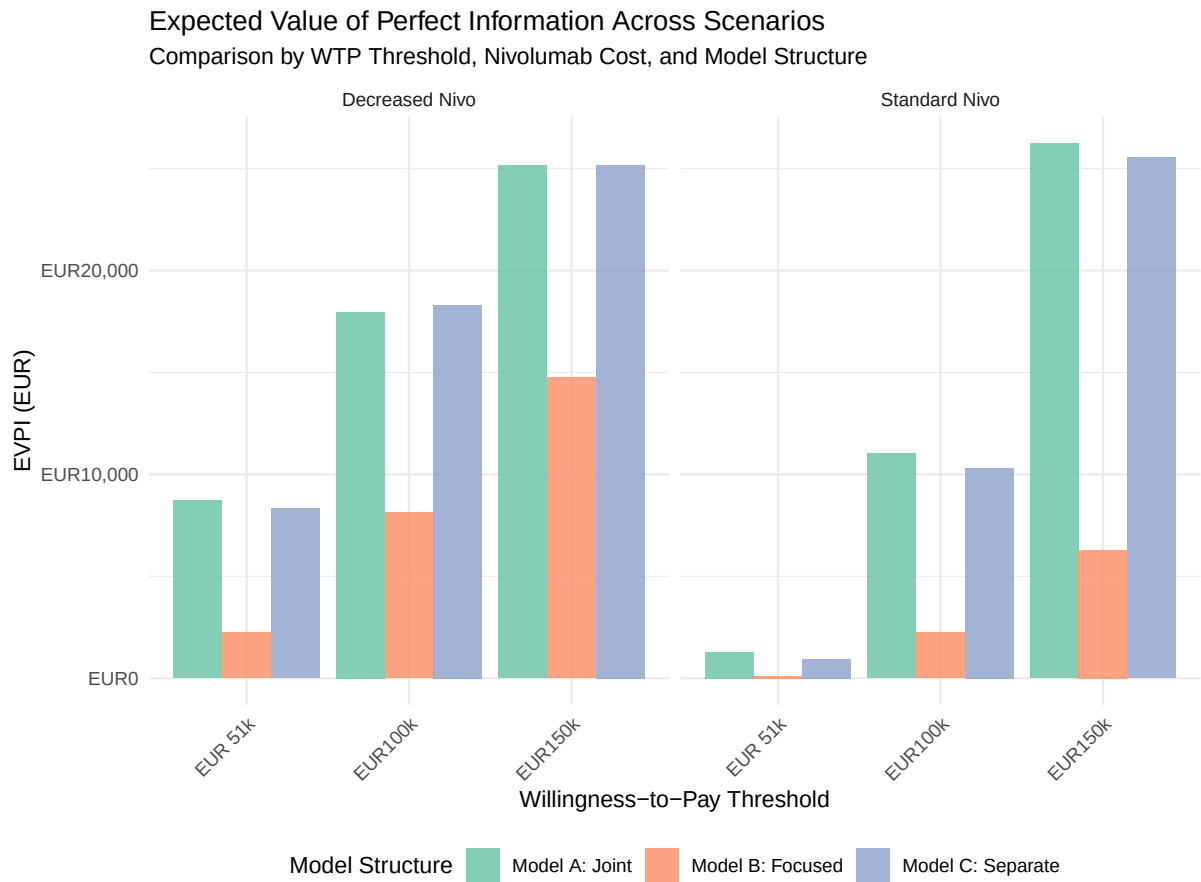


Figure 8: Expected Value of Perfect Information across scenarios and model structures

10.2 EVPPI Parameter Heatmap by Model Structure

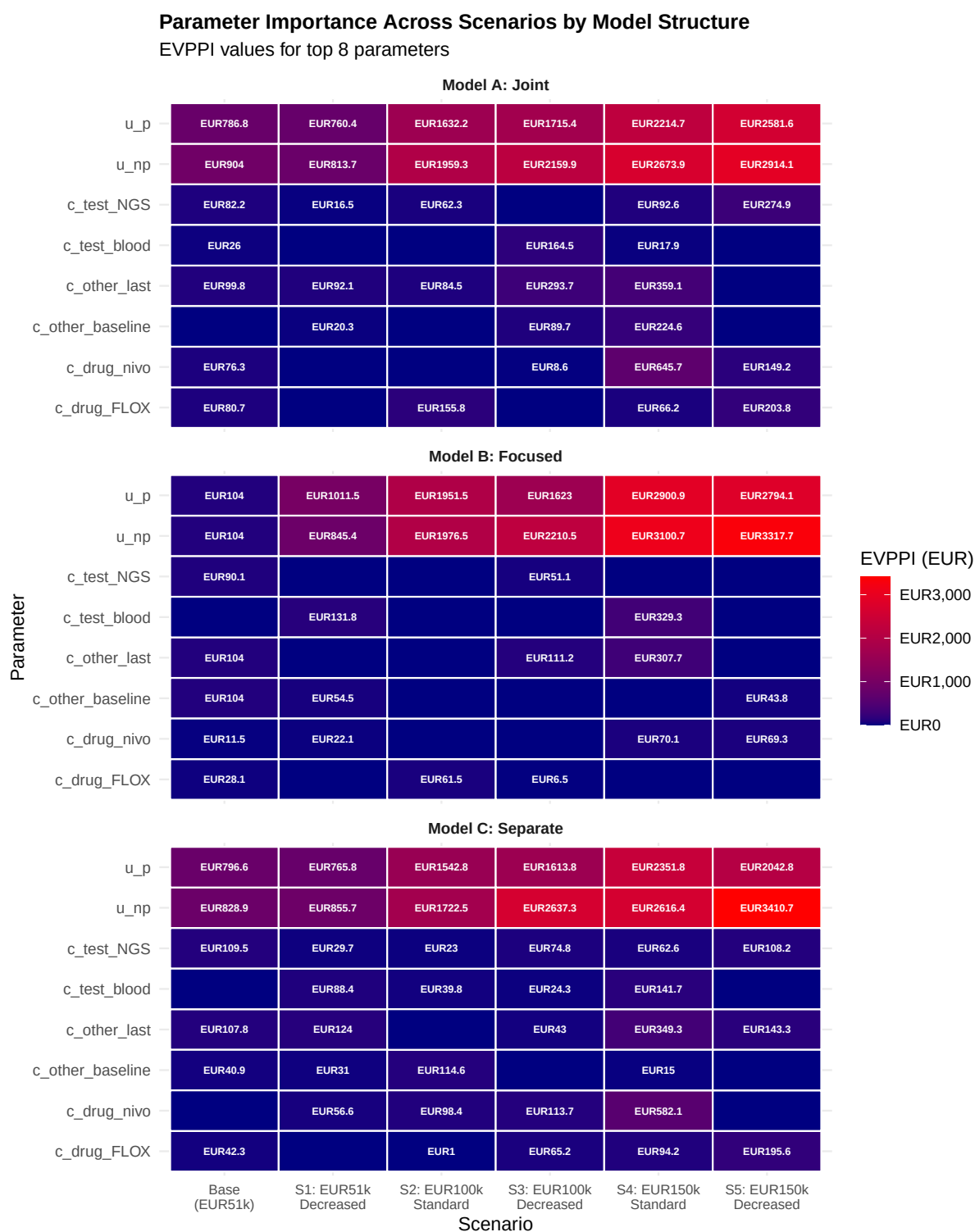


Figure 9: Parameter importance across scenarios by model structure (top 8 parameters)

10.3 Summary Comparison Table

Table 15: Summary of EVPI and top EVPPI by model structure and scenario

Model	Scenario	WTP	EVPI	Top Parameter	Max EVPPI	% of EVPI
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Model A: Joint	Base Case	EUR 51,000	EUR1291.45	u_np	EUR903.95	70.0
Model A: Joint	Decreased Nivolumab Cost	EUR 51,000	EUR8728.45	u_np	EUR813.71	9.3
Model A: Joint	WTP €100,000	EUR100,000	EUR11036.3	u_np	EUR1959.34	17.8
Model A: Joint	WTP €100,000 + Decreased Nivolumab	EUR100,000	EUR17969.61	u_np	EUR2159.89	12.0
Model A: Joint	WTP €150,000	EUR150,000	EUR26232	u_np	EUR2673.87	10.2
Model A: Joint	WTP €150,000 + Decreased Nivolumab	EUR150,000	EUR25162.41	u_np	EUR2914.08	11.6
Model B: Focused	Base Case	EUR 51,000	EUR103.98	u_np	EUR103.98	100.0
Model B: Focused	Decreased Nivolumab Cost	EUR 51,000	EUR2235.35	u_p	EUR1011.52	45.3
Model B: Focused	WTP €100,000	EUR100,000	EUR2233.74	u_np	EUR1976.48	88.5
Model B: Focused	WTP €100,000 + Decreased Nivolumab	EUR100,000	EUR8134.99	u_np	EUR2210.46	27.2
Model B: Focused	WTP €150,000	EUR150,000	EUR6287.53	u_np	EUR3100.7	49.3
Model B: Focused	WTP €150,000 + Decreased Nivolumab	EUR150,000	EUR14756.5	u_np	EUR3317.74	22.5
Model C: Separate	Base Case	EUR 51,000	EUR940.43	u_np	EUR828.86	88.1
Model C: Separate	Decreased Nivolumab Cost	EUR 51,000	EUR8326.24	u_np	EUR855.66	10.3
Model C: Separate	WTP €100,000	EUR100,000	EUR10320.09	u_np	EUR1722.5	16.7
Model C: Separate	WTP €100,000 + Decreased Nivolumab	EUR100,000	EUR18313.84	u_np	EUR2637.27	14.4
Model C: Separate	WTP €150,000	EUR150,000	EUR25542.16	u_np	EUR2616.38	10.2
Model C: Separate	WTP €150,000 + Decreased Nivolumab	EUR150,000	EUR25182.37	u_np	EUR3410.67	13.5

10.4 Key Findings

10.4.1 Model Structure Robustness

Parameter importance is consistent across model structures. The parameter ‘u_np’ is the most important driver of value of information across all three model specifications. This suggests that conclusions about research priorities are robust to the choice of model structure.

10.4.2 Scenario with Greatest Decision Uncertainty

- **Model A: Joint:** WTP €150,000 (EVPI = EUR26232)
- **Model B: Focused:** WTP €150,000 + Decreased Nivolumab (EVPI = EUR14756.5)
- **Model C: Separate:** WTP €150,000 (EVPI = EUR25542.16)

10.4.3 Most Important Parameter Across All Scenarios

u_np (Total EVPPI = EUR35050.96)

10.4.4 General Observations

Impact of WTP Threshold: Higher WTP thresholds generally increase decision uncertainty (EVPI), creating greater value for additional research to reduce parameter uncertainty.

Impact of Nivolumab Cost: Decreased nivolumab costs shift the cost-effectiveness landscape and may change which parameters drive decision uncertainty.

Impact of Model Structure: Comparing EVPI and EVPPI across model structures helps identify whether value of information conclusions are robust to structural uncertainty in the survival model specification.

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